

Users > moshika > Desktop > lab >  BFS.pl

```
1    % --- Graph Edges ---
2    edge(a, b).
3    edge(a, c).
4    edge(b, d).
5    edge(b, e).
6    edge(c, f).
7    edge(c, g).
8    edge(e, h).
9
10   % --- BFS Driver ---
11   bfs(Start, Goal, Path) :-
12       bfs_search([Start], Goal, RevPath),
13       reverse(RevPath, Path).
14
15   % --- BFS Search Logic ---
16   bfs_search([Goal | RestPath] | _, Goal, [Goal | RestPath]) :- !.
17
18   bfs_search([Node | Path] | Queue, Goal, Result) :-
19       findall([NextNode, Node | Path],
20           (edge(Node, NextNode), \+ member(NextNode, [Node | Path])),
21           Children),
22       append(Queue, Children, NewQueue),
23       bfs_search(NewQueue, Goal, Result).
24
```

```
moshika@Moshikas-MacBook-Air ~ % cd /Users/moshika/Desktop/lab  
swipl BFS.pl
```

```
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```
For online help and background, visit https://www.swi-prolog.org  
For built-in help, use ?- help(Topic). or ?- apropos(Word).
```

```
[?- bfs(a, h, Path).  
Path = [a, b, e, h].
```